ICT/OLevels

English Medium

Learn ICT with Shawn sir.

Grade 10

5 Logic Gates

What are Logic Gates

Draw 3 Basic Logic Gates and 2 Combined Logic Gates

AND

OR

NOT

NAND

NOR

Draw the following Logic Gates using the Algebraic equation

$(C . D)$

$\overline{(A + B)} + (B . C)$

$(A + C) + C$

$(A + B) + (A . B)$

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C . (A + B)

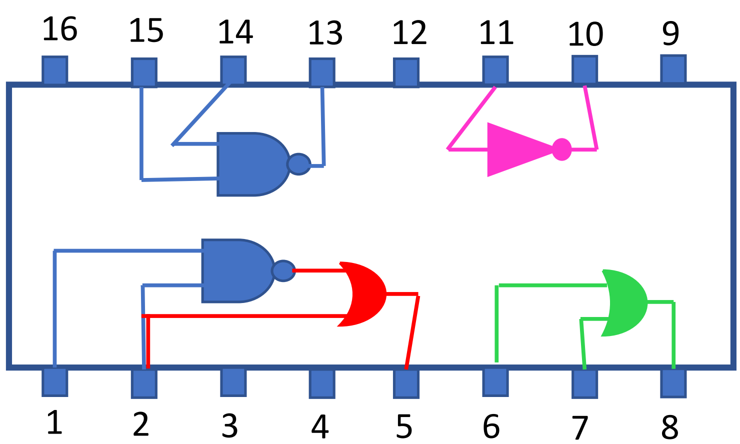
$\overline{C}$ + (A + B)

C + (A . B)

$\overline{C}$ + (A . B)

The following IC (Integrated Circuit) diagram was taken from a FM Radio Diagram tuner. Unlike Microprocessors in Computers & advanced electronics, IC's were primarily used in consumer electronics since the 1970's.

Find the output of the given pin when the correct input voltage is supplied to their respective pins.



- When pin1 = 1 and pin2 = 0, what will be the value of pin5
- When pin6 = 1 and pin7 = 1, what will be the value of pin8
- When pin14 = 0 and pin15 = 1, what will be the value of pin13
- When pin11 = 1, what will be the value of pin10
- What's the value of pin15, when pin14 = 1 and pin13 = 0
- What's the value of pin10, when pin11 is 0
- What's the value of pin1, when pin2 = 0 and pin5 = 1

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Real-World Application

An excellent security system should activate an alarm only when:

- The front-door is open (A = 1)
- The back-door is open (B = 1)
- The window is open (C = 1)

Which logic gate should be used
(Draw the logic gate with the A B inputs)

Real-World Application

A technical student built an Arduino based temperature Fan Regulation system for his class. The Fan will activate only when:

- The Fan switch is ON (A = 1)
- The windows is opened (B = 0)
- The temperature is above 25"C (C = 1)

Which combination of logic gate should be used (draw the logic gates with all inputs)

Real-World Application

School Computer Login System:

A student can access a computer only when:

- The correct Username is entered (A = 1)
- The correct Password is entered (B = 1)

Which logic gate best represents this situation?
Draw a TRUTH table for this.

Real-World Application

School Science Laboratory

A chemical cabinet should unlock only when:

- The teacher's key card is detected (A = 1)
- The fingerprint scan is successful (B = 1)
- Time is between 7AM and 9PM (C = 1)

However, if anyone is false, the cabinet must remain locked. Which combinational logic gates should control the lock?

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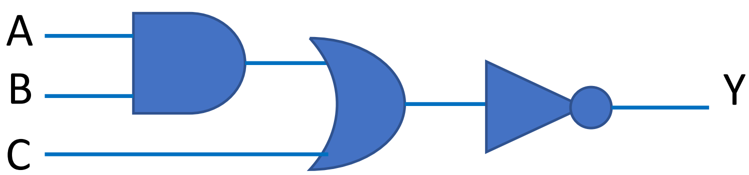
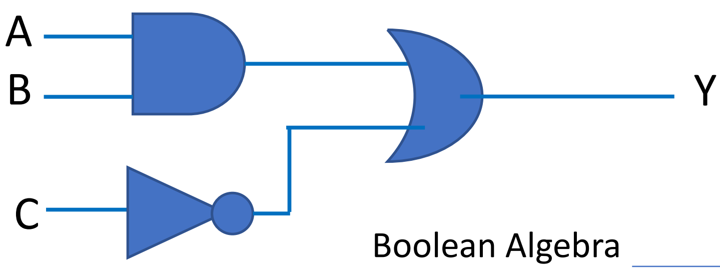
Logic Gates to algebraic equation

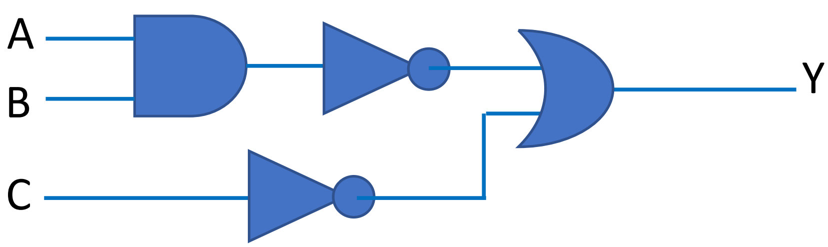
A mega-factory in Japan is constructing a robot assembly and requires the following Logic Gates to be documented in their computer system. As of now, the photo-lithography design system can only accept Boolean expressions. Convert the following Logic Gate diagrams to Boolean algebra.



Boolean Algebra ______________________________



Boolean Algebra ______________________________



Boolean Algebra ______________________________

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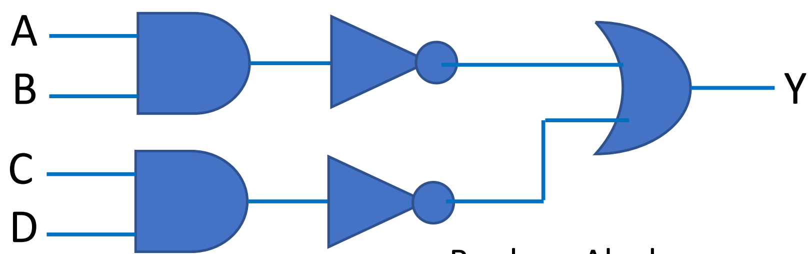
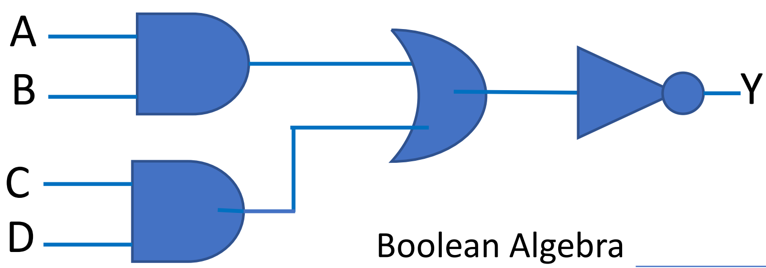
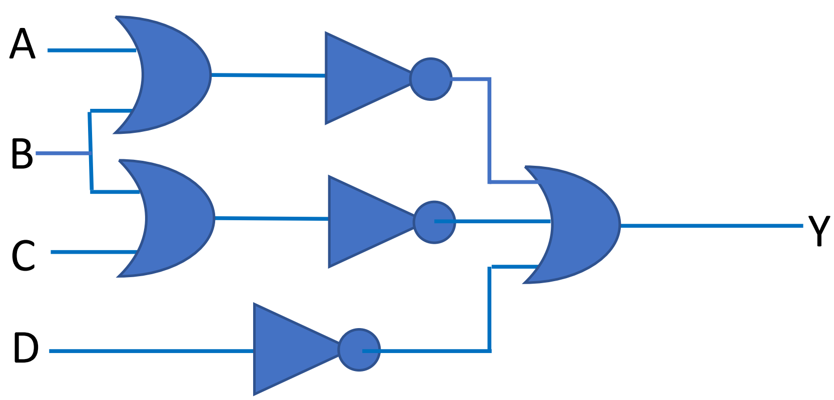
Logic Gates to algebraic equation
continued…



Boolean Algebra ______________________



Boolean Algebra ______________________



Boolean Algebra ______________________

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